



Session 12A2: Matrices and Vectors — The Algebra of Space

Phase 2 — Classical Techniques | 80 min

Part A: Matrices — Rectangular Arrays That Transform Space

Example 1: What a Matrix Is — A Grid of Numbers

A matrix is a rectangular array of numbers arranged in rows and columns.

$A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$ is a 2×2 matrix (2 rows, 2 columns).

The entry in row i , column j is written as a_{ij} :

$a_{11} = 2, a_{12} = 3, a_{21} = 1, a_{22} = 4$.

Matrices can be any size: $3 \times 2, 4 \times 4, 1 \times 5$, etc.

Example 2: Adding and Scaling Matrices — Entry by Entry

Add two matrices of the same size: add corresponding entries.

$$\begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix} + \begin{pmatrix} 5 & -1 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 7 & 2 \\ 1 & 6 \end{pmatrix}.$$

Multiply a matrix by a scalar: multiply every entry by that number.

$$3 \cdot \begin{pmatrix} 2 & -1 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} 6 & -3 \\ 0 & 12 \end{pmatrix}.$$

Both operations work entry-by-entry — no special rules needed.

Example 3: Matrix Multiplication — Row Meets Column

To multiply $A \times B$, each entry (i, j) of the result comes from pairing row i of A with column j of B .

$$A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}, B = \begin{pmatrix} 5 & 1 \\ 0 & 2 \end{pmatrix}.$$

$$A \times B = \begin{pmatrix} 2 \cdot 5 + 3 \cdot 0 & 2 \cdot 1 + 3 \cdot 2 \\ 1 \cdot 5 + 4 \cdot 0 & 1 \cdot 1 + 4 \cdot 2 \end{pmatrix} = \begin{pmatrix} 10 & 8 \\ 5 & 9 \end{pmatrix}.$$

The size rule: To multiply A ($m \times n$) by B ($n \times p$), the inner dimensions must match ($n = n$). The result is $m \times p$.

Matrix multiplication is not commutative: $A \times B \neq B \times A$ in general.

$$\text{Check: } B \times A = \begin{pmatrix} 5 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix} = \begin{pmatrix} 11 & 19 \\ 2 & 8 \end{pmatrix} \neq A \times B.$$

Example 4: The Identity Matrix and the Inverse

The **identity matrix** I acts like the number 1: multiplying by I leaves a matrix unchanged.

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \text{ Check: } A \times I = I \times A = A.$$

The **inverse** of A , written A^{-1} , satisfies $A \times A^{-1} = A^{-1} \times A = I$.

$$\text{For a } 2 \times 2 \text{ matrix } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}:$$

$$A^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \text{ provided } ad - bc \neq 0.$$

Example: $A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}.$

$$ad - bc = 2 \cdot 4 - 3 \cdot 1 = 8 - 3 = 5.$$

$$A^{-1} = \frac{1}{5} \begin{pmatrix} 4 & -3 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 0.8 & -0.6 \\ -0.2 & 0.4 \end{pmatrix}.$$

Check: $A \times A^{-1} = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 0.8 & -0.6 \\ -0.2 & 0.4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$ It works.

The number $ad - bc$ is called the **determinant**, written $\det(A)$. If $\det(A) = 0$, the matrix has no inverse — it is **singular**.

Example 5: Determinant — The Area Scaling Factor

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\det(A) = ad - bc$.

Geometric meaning: $|\det(A)|$ is the factor by which the matrix scales area.

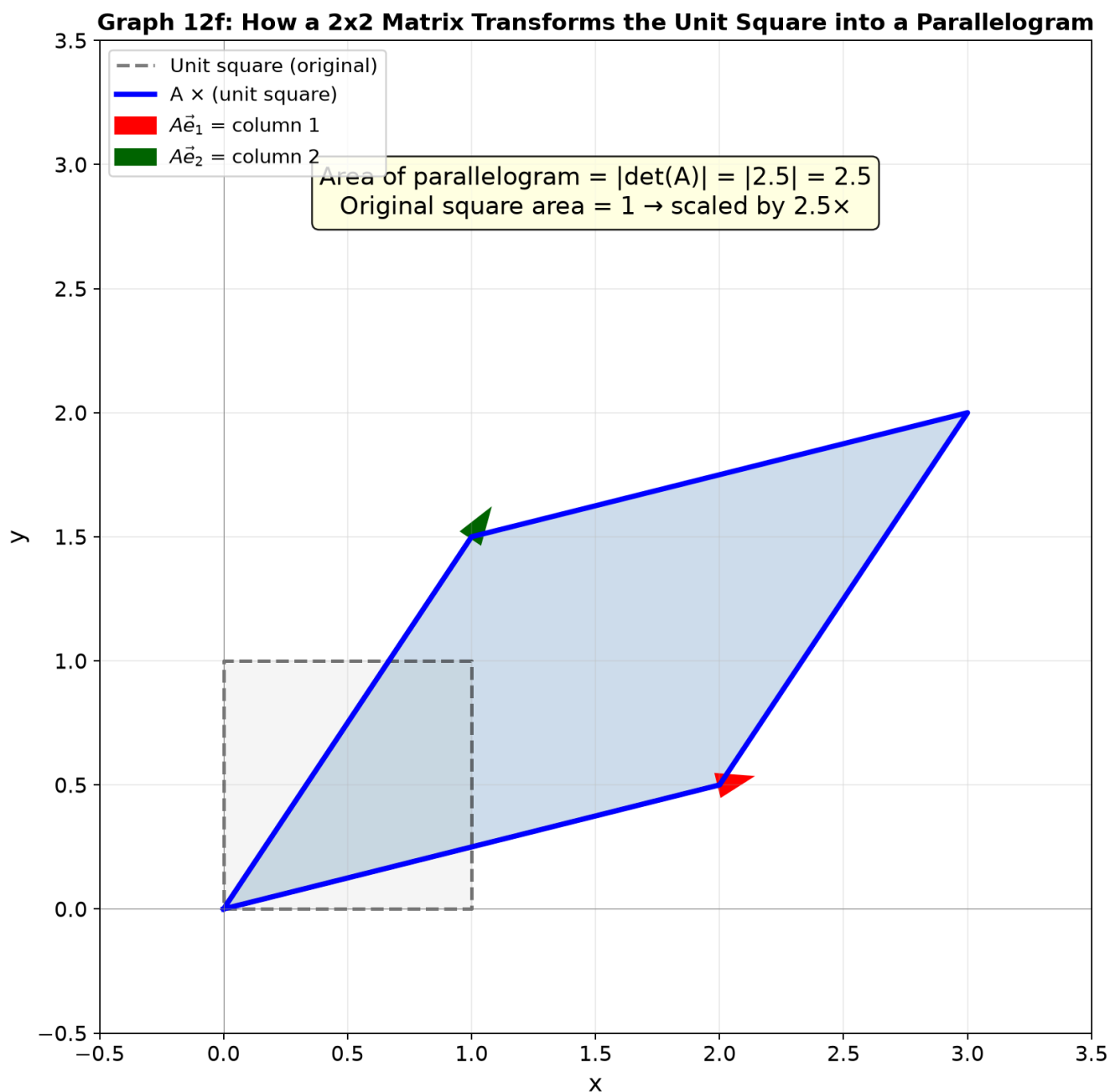
$A = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$ stretches x by 3 and y by 2.

$\det(A) = 6$. The unit square (area 1) becomes a 3×2 rectangle (area 6).

$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ rotates by 90° clockwise.

$\det(A) = 0 \cdot 0 - 1 \cdot (-1) = 1$. Rotation preserves area.

If $\det(A) = 0$, the transformation collapses the plane onto a line or a point — area becomes 0.



Example 6: Solving Linear Systems with Matrices

The system $\begin{cases} 2x + 3y = 5 \\ x + 4y = 6 \end{cases}$ can be written as $A\vec{x} = \vec{b}$:

$$\begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}.$$

Solve by multiplying both sides by A^{-1} : $\vec{x} = A^{-1}\vec{b}$.

$$A^{-1} = \frac{1}{5} \begin{pmatrix} 4 & -3 \\ -1 & 2 \end{pmatrix}.$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 4 & -3 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 20 - 18 \\ -5 + 12 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 2 \\ 7 \end{pmatrix} = \begin{pmatrix} 0.4 \\ 1.4 \end{pmatrix}.$$

Check: $2(0.4) + 3(1.4) = 0.8 + 4.2 = 5 \checkmark$. $0.4 + 4(1.4) = 0.4 + 5.6 = 6 \checkmark$.

Example 7: Matrix Powers — Composing Transformations

Applying A twice means computing $A^2 = A \times A$.

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \text{ (rotation by } 90^\circ\text{)}.$$

$$A^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \text{ (rotation by } 180^\circ = -I\text{)}.$$

$$A^3 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \text{ (rotation by } 270^\circ\text{)}.$$

$$A^4 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \text{ (full rotation — back to start).}$$

Matrices can represent geometric transformations — rotation, reflection, scaling, shearing — and matrix powers compose those transformations.

Up to here: Matrices are grids. Add and scale entry-by-entry. Multiply via row $\times$ column.

The inverse undoes a matrix: $A^{-1}A = I$. Determinant = area scaling factor.

Use $A^{-1}\vec{b}$ to solve linear systems. Matrix powers compose transformations.

Part B: Vectors — Direction and Magnitude in Space

Example 8: Vector Basics — Components and Magnitude

A vector is a quantity with both direction and length. In 2D: $\vec{a} = (a_1, a_2)$. In 3D: $\vec{a} = (a_1, a_2, a_3)$

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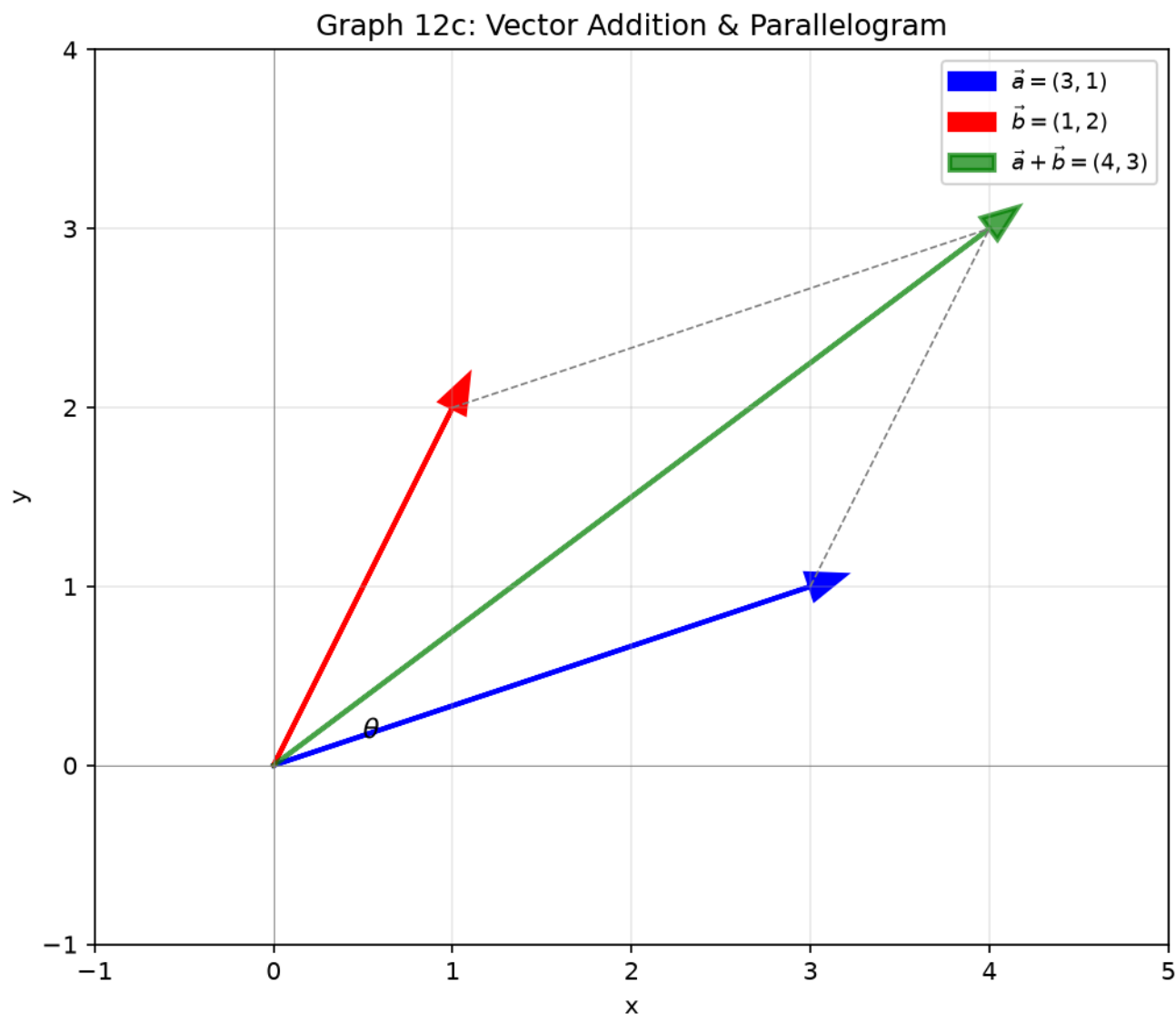
Magnitude (length): $|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$.

$$\vec{a} = (3, 4). |\vec{a}| = \sqrt{9 + 16} = 5.$$

Unit vector (length 1, same direction): $\hat{a} = \frac{\vec{a}}{|\vec{a}|} = (\frac{3}{5}, \frac{4}{5})$.

Add vectors component-wise: $\vec{a} + \vec{b} = (a_1 + b_1, a_2 + b_2)$.

Scale a vector: $k\vec{a} = (ka_1, ka_2)$.



Example 9: Dot Product — Angle and Perpendicular Test

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3 = |\vec{a}||\vec{b}| \cos \theta.$$

The dot product gives the cosine of the angle between two vectors.

$$(3, 4) \cdot (1, 2) = 3 \cdot 1 + 4 \cdot 2 = 3 + 8 = 11.$$

$$\cos \theta = \frac{11}{5 \cdot \sqrt{5}} = \frac{11}{5\sqrt{5}}. \theta \approx 10.3^\circ \text{ — nearly parallel.}$$

Dot product = 0 $\rightarrow$ perpendicular.

$$(3, 4) \cdot (-4, 3) = -12 + 12 = 0. \text{ The vectors are at right angles.}$$

Dot product sign:

- Positive $\rightarrow$ angle $< 90^\circ$ (acute).
- Zero $\rightarrow$ exactly 90° (perpendicular).
- Negative $\rightarrow$ angle $> 90^\circ$ (obtuse).

Example 10: Vector Projection — The Shadow of One Vector on Another

Scalar projection (the length of the shadow): $\text{comp}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$.

Vector projection (the shadow as a vector): $\text{proj}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \vec{b}$.

Example: $\vec{a} = (3, 4)$, $\vec{b} = (1, 0)$ (the x -axis).

Scalar projection = $\frac{3 \cdot 1 + 4 \cdot 0}{1} = 3$. The shadow of $(3, 4)$ on the x -axis has length 3.

Vector projection = $3 \cdot (1, 0) = (3, 0)$. The shadow as a vector is $(3, 0)$.

The projection formula decomposes any vector into a part parallel to $\vec{b}$ and a part perpendicular to $\vec{b}$.

Example 11: Cross Product — 3D Only, Produces a Perpendicular Vector

The cross product $\vec{a} \times \vec{b}$ gives a vector perpendicular to both $\vec{a}$ and $\vec{b}$.

$$\vec{a} \times \vec{b} = (a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1).$$

Example: $(1, 0, 0) \times (0, 1, 0) = (0, 0, 1)$.

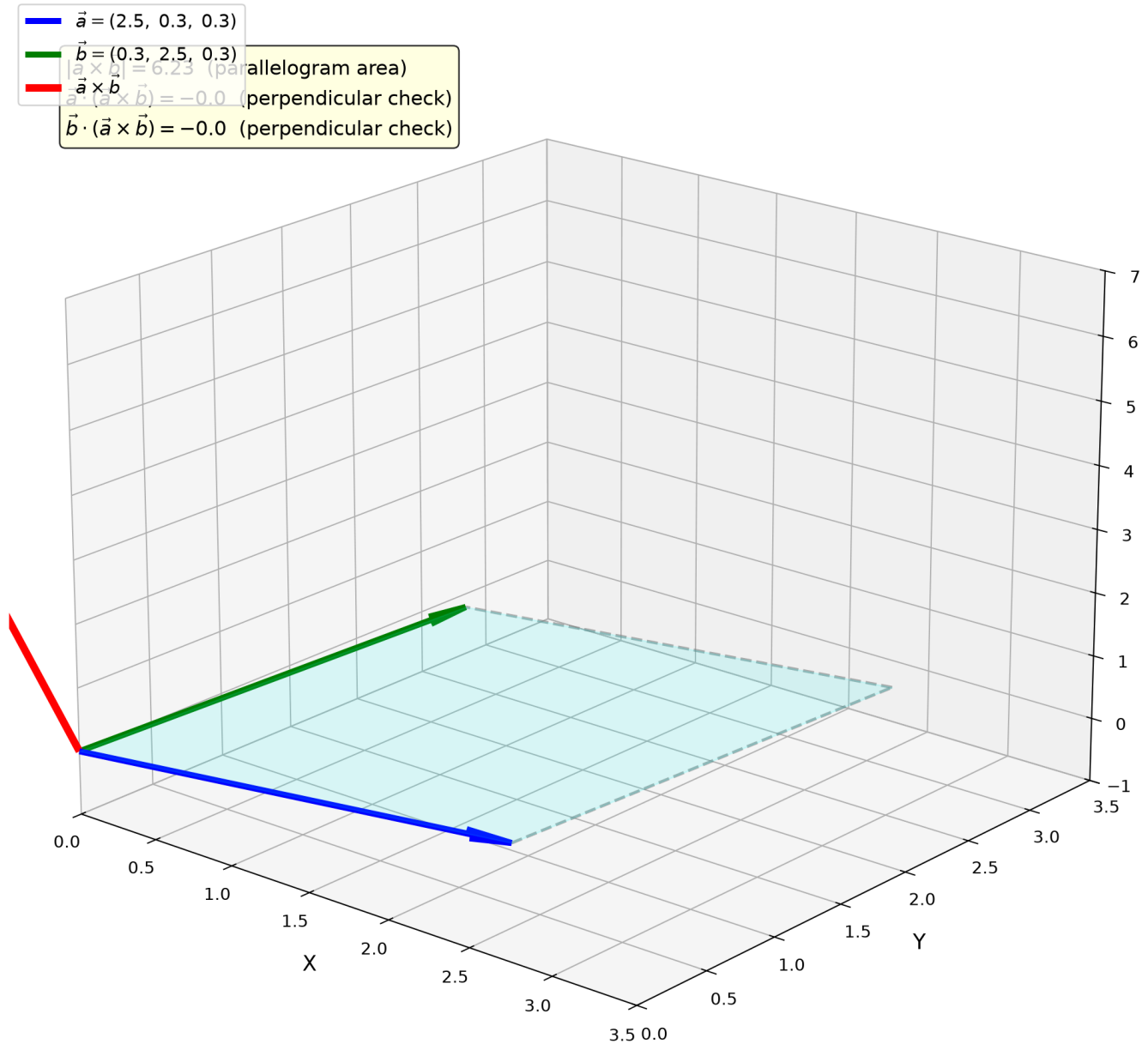
The x -axis crossed with the y -axis produces the z -axis — right-hand rule.

Magnitude of cross product: $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin \theta$.

This equals the area of the parallelogram formed by $\vec{a}$ and $\vec{b}$.

Cross product = 0 $\rightarrow$ parallel vectors (since $\sin \theta = 0$).

Graph 12h: Cross Product — A Vector Perpendicular to Both a and b



Example 12: Using Vectors for Geometry — Area and Distance

Triangle area (3D): $\frac{1}{2}|\vec{AB} \times \vec{AC}|$.

Triangle area (2D): $\frac{1}{2}|a_1b_2 - a_2b_1|$ where (a_1, a_2) and (b_1, b_2) are two side vectors.

$$\vec{AB} = (3, 1), \vec{AC} = (1, 4). \text{ Area} = \frac{1}{2}|3 \cdot 4 - 1 \cdot 1| = \frac{1}{2} \cdot 11 = \frac{11}{2}.$$

Distance from a point to a line (3D): line through P with direction $\vec{d}$, point Q .

$$d = \frac{|\vec{d} \times (\vec{Q} - \vec{P})|}{|\vec{d}|}.$$

The cross product appears whenever you need a perpendicular direction or an area.

Up to here: Vectors have magnitude and direction. Dot product $\rightarrow$ angle. Cross product $\rightarrow$ perpendicular direction + area.

Projection $\rightarrow$ shadow along a direction. Vectors solve geometry without coordinate geometry formulas.

Part C: How Matrices and Vectors Work Together

Example 13: Matrices as Transformations of Vectors

A matrix A takes a vector $\vec{x}$ as input and produces a new vector $A\vec{x}$ as output. This is a **linear transformation** — it preserves addition and scalar multiplication.

The columns of A tell you exactly where the standard basis vectors go:

- Column 1 = $A\vec{e}_1$ (image of the x -axis direction).
- Column 2 = $A\vec{e}_2$ (image of the y -axis direction).

Example: $A = \begin{pmatrix} 2 & 1 \\ 0.5 & 1.5 \end{pmatrix}$ maps $\vec{e}_1 = (1, 0)$ to $(2, 0.5)$ and $\vec{e}_2 = (0, 1)$ to $(1, 1.5)$. The unit square becomes the parallelogram spanned by these two column vectors.

Rotation matrix: $R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ rotates every vector by θ counterclockwise.

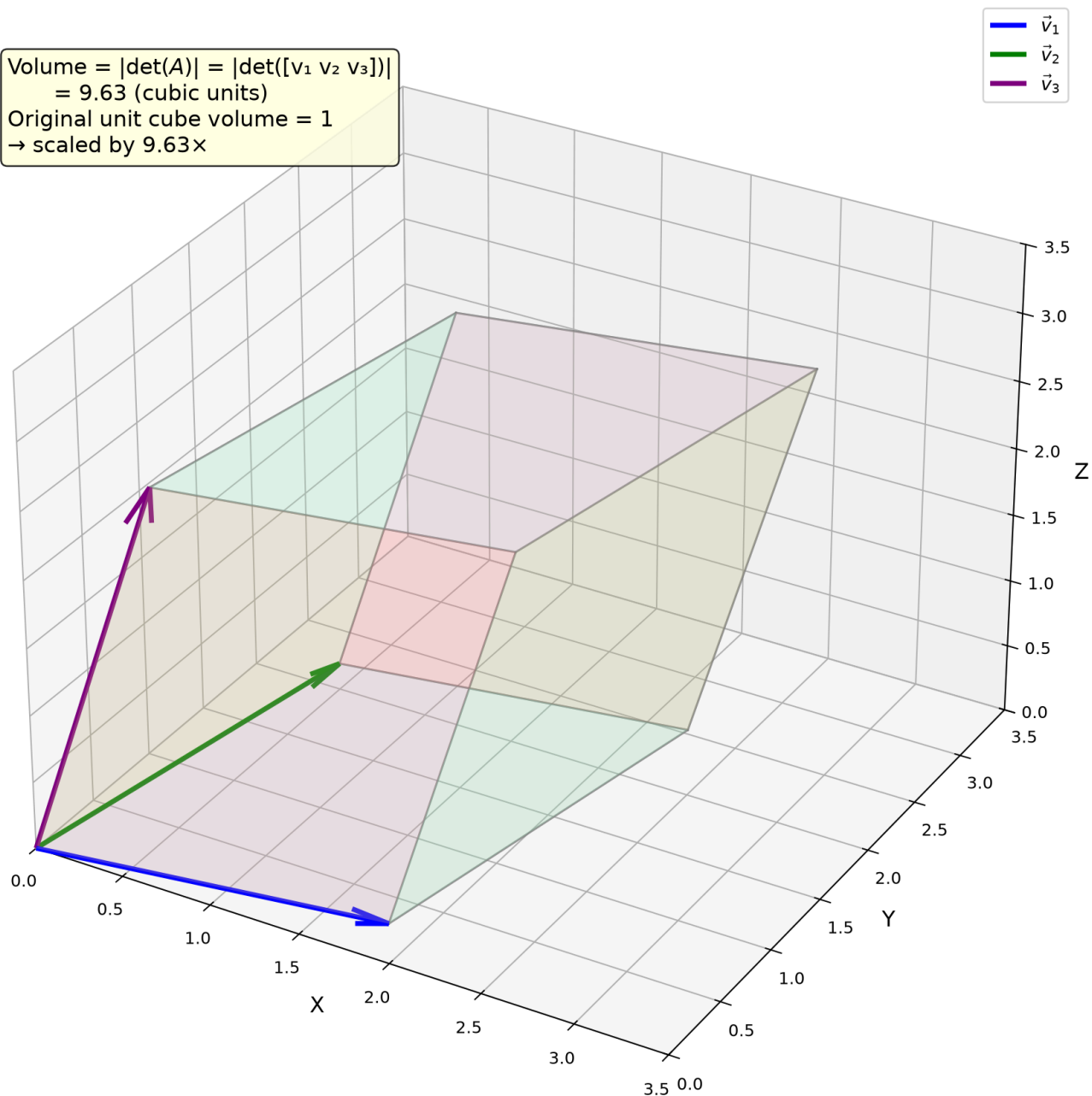
Visual Interlude: From 2D to 3D to n D

Dimension 2 — Area. In 2D, the determinant scales area. A 2×2 matrix turns the unit square into a parallelogram. Area = $|\det(A)|$.

Dimension 3 — Volume. In 3D, the determinant scales volume. A 3×3 matrix turns the unit cube into a parallelepiped.

Graph 12i: Determinant = Volume of the Parallelepiped (3D)

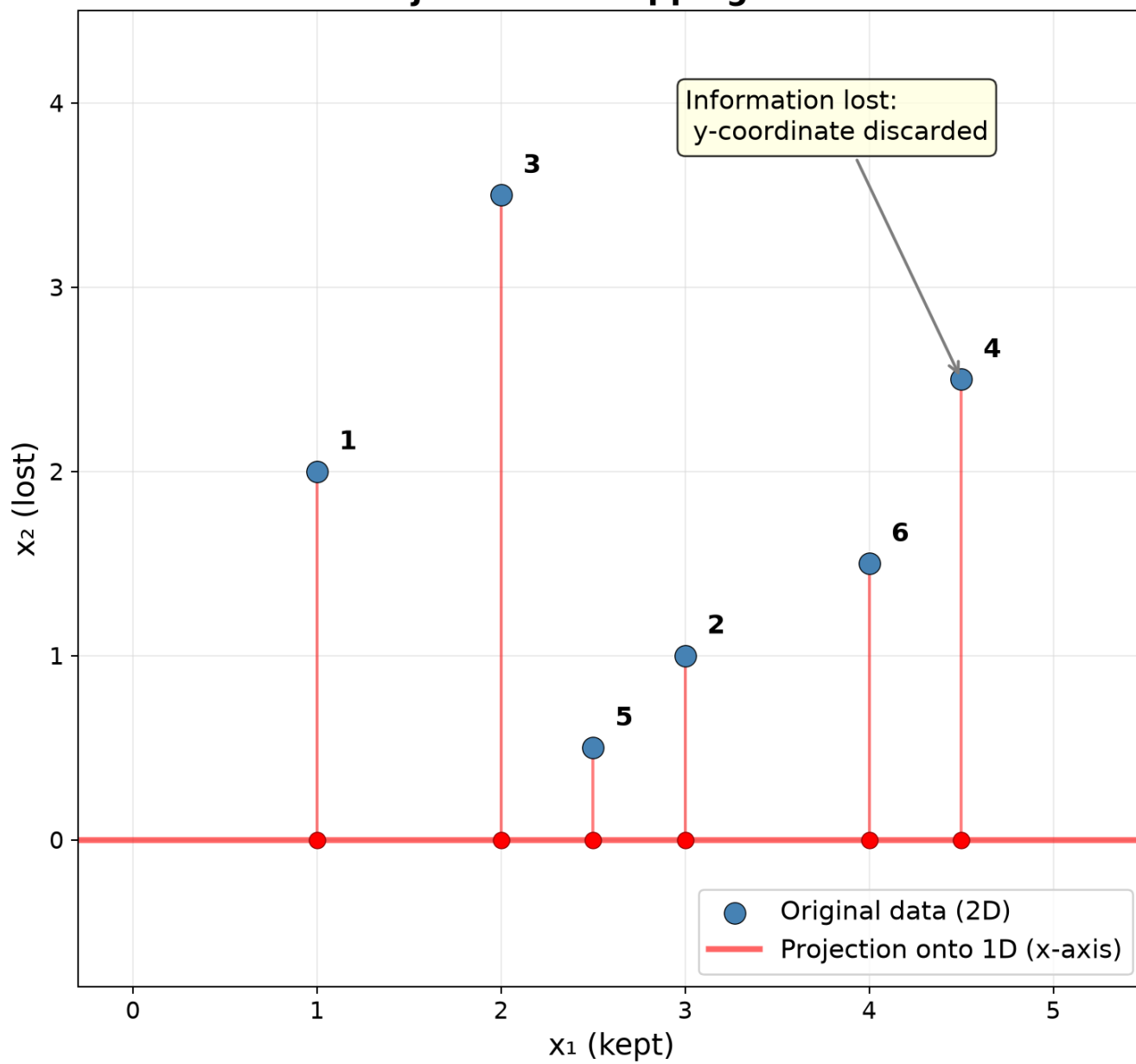
Volume = $|\det(A)| = |\det([v_1 \ v_2 \ v_3])|$
= 9.63 (cubic units)
Original unit cube volume = 1
→ scaled by 9.63×



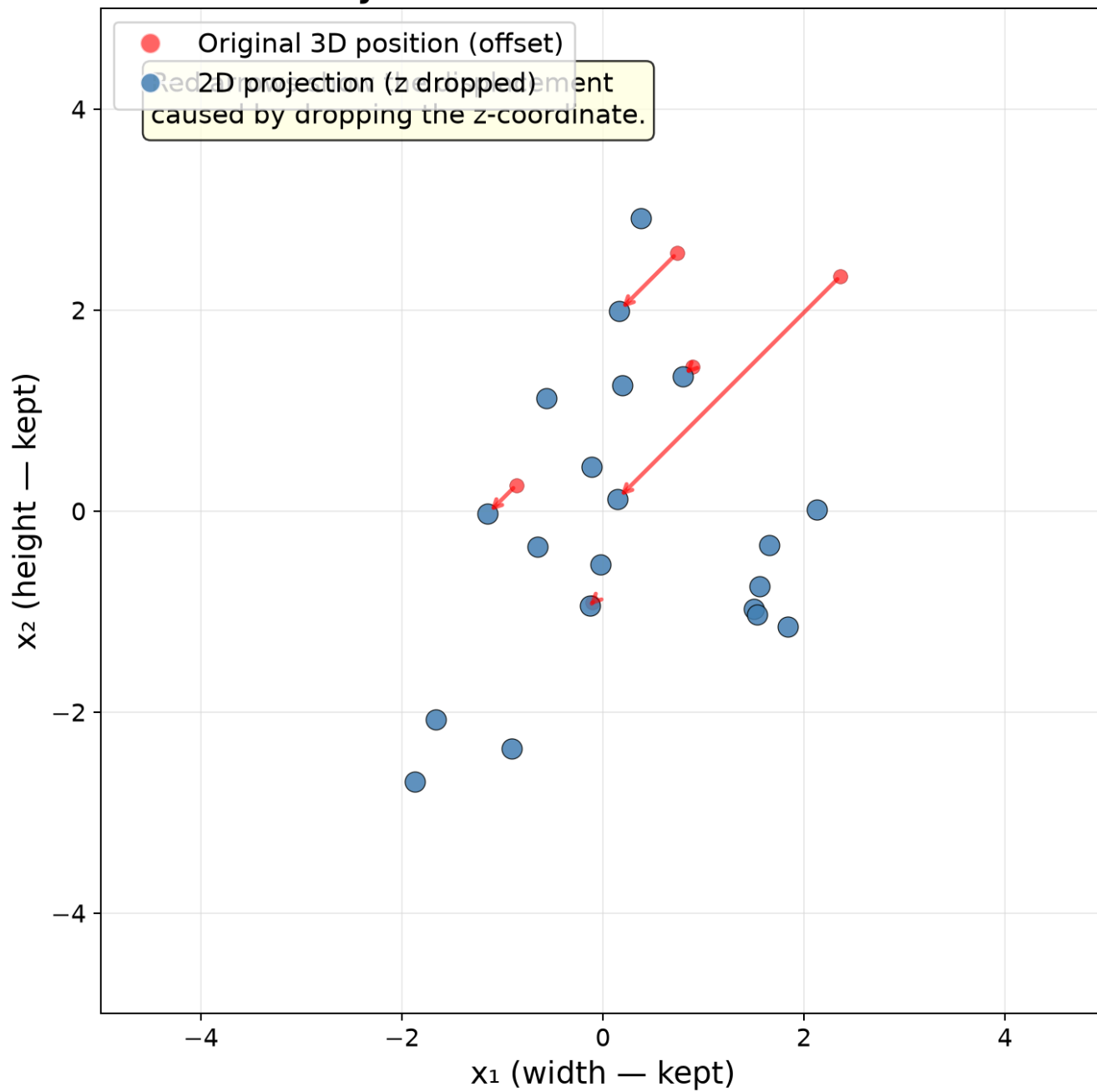
Beyond 3D — Hyper-volume. In 4D and higher, we lose direct visual intuition. But the determinant still scales "hyper-volume" in exactly the same way.

How projection works — reducing dimension step by step:

2D → 1D Projection — Dropping One Dimension



3D → 2D Projection — The z-dimension is Flattened



Dimensionality Reduction Pipeline

The diagram illustrates a pipeline with three nested rectangles representing different stages of data representation:

- Outermost Rectangle (Pink):** Labeled R^n (Input). It represents the original high-dimensional data.
- Middle Rectangle (Green):** Labeled $A \in \mathbb{R}^{m \times n}$ (Encoding). It represents the data after an initial encoding step.
- Innermost Rectangle (Purple):** Labeled $A^* \in \mathbb{R}^{k \times n}$ (Essential Information). It represents the data after a more refined encoding step, capturing the essential information.

A vertical arrow points upwards through the center of the nested rectangles, indicating the flow of the pipeline from input to the final essential information stage.

Annotation: A 4D→2D projection matrix is just a 2×4 matrix. It loses information — but the essential structure.

A 4D→2D projection matrix is just a 2×4 matrix.
It loses information — but the essential structure
(clusters, directions, relative distances) often survives.

A 4D→2D projection (like PCA or t-SNE in machine learning) is just a 2×4 matrix that collapses 4 dimensions onto a flat screen while preserving as much structure as possible. The determinant in higher dimensions still measures the volume-scaling factor — we just cannot draw it directly.

The unifying idea: In every dimension, a matrix is a transformation of space. The determinant measures how much the transformation scales volume. Vectors are the objects being transformed. The dot product, cross product, and projection are tools for measuring relationships between vectors — angles, areas, shadows — in any dimension.

Common Mistakes

Mistake 1: Forgetting that $AB \neq BA$ for matrices

Wrong path: "Matrix multiplication works like regular multiplication — order does not matter."

Why wrong: AB and BA are usually different matrices. Matrix multiplication is not commutative.

Right path: Always check the order. A times B means apply B first, then A (when thinking about transformations).

Mistake 2: Dot product negative $\rightarrow$ acute angle

Wrong path: "A negative dot product means the angle is less than 90° ."

Why wrong: $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta$. If the dot product is negative, $\cos \theta$ is negative, so $\theta > 90^\circ$ (obtuse).

Right path: Dot product sign = $\cos \theta$ sign. Positive $\rightarrow$ acute. Zero $\rightarrow$ right angle. Negative $\rightarrow$ obtuse.

Mistake 3: Cross product is commutative

Wrong path: " $\vec{a} \times \vec{b} = \vec{b} \times \vec{a}$."

Why wrong: $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$. The cross product is anti-commutative — swapping the order flips the sign.

Right path: $\vec{a} \times \vec{b}$ points one way (right-hand rule). $\vec{b} \times \vec{a}$ points the opposite way.

Mistake 4: Trying to invert a singular matrix

Wrong path: "I'll find A^{-1} using the formula... oh wait, dividing by zero?"

Why wrong: If $\det(A) = 0$, the matrix has no inverse. The transformation collapses space and cannot be undone.

Right path: Check $\det(A)$ first. If it is 0, the system either has no solution or infinitely many.

What We Just Did

- (1) Matrices – add entry-by-entry. Multiply: row $i \times$ column j .
Identity I acts like 1. Inverse A^{-1} undoes A .
Determinant = area/volume scaling factor.
Solve $A\vec{x} = \vec{b}$ via $\vec{x} = A^{-1}\vec{b}$.
 - (2) Vectors – magnitude = $|\vec{a}|$. Dot product $\rightarrow \cos \theta \rightarrow$ angle.
Cross product $\rightarrow$ perpendicular vector + parallelogram area.
Projection $\rightarrow$ shadow of one vector onto another.
Triangle area = $\frac{1}{2}|\vec{AB} \times \vec{AC}|$.
 - (3) Connection – matrix columns = images of basis vectors.
A 2×2 matrix maps the unit square to a parallelogram (area = $|\det|$).
A 3×3 matrix maps the unit cube to a parallelepiped (volume = $|\det|$).
In nD , the determinant scales hyper-volume. Projection matrices reduce dimension.
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Practice 1

A 2×2 matrix A has determinant 0. What does this tell you geometrically about the transformation A represents? Give a concrete example of such a matrix and describe what it does to the unit square.

→ Reference: **Example 5**

Solutions: [Solutions](#)

Practice 2

Solve the system using matrix inversion:

$$\begin{cases} 3x - 2y = 7 \\ x + 4y = 5 \end{cases}$$

→ Reference: **Example 6**

Solutions: [Solutions](#)

Practice 3

$\vec{a} = (2, -1, 3)$, $\vec{b} = (1, 4, -2)$. Find the dot product and the cross product. Verify that the cross product is perpendicular to both $\vec{a}$ and $\vec{b}$.

→ Reference: **Example 9, 11**

Solutions: [Solutions](#)

Practice 4

Find the area of the triangle with vertices $A(0, 0, 0)$, $B(3, 1, 0)$, $C(1, 4, 0)$ using the cross product.

→ Reference: **Example 12**

Solutions: [Solutions](#)

Practice 5: Composition

Invent a 2×2 matrix that represents a reflection across the x -axis, and another that represents a 90° counterclockwise rotation. Multiply them in both orders and describe the geometric meaning of each product.

→ Reference: **Example 7**

Solutions: [Solutions](#)

Practice 6

Find the scalar and vector projections of $\vec{a} = (5, 12)$ onto $\vec{b} = (3, 4)$.

→ Reference: **Example 10**

Solutions: [Solutions](#)

Practice 7

For $A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$, compute A^{-1} and verify $AA^{-1} = I$.

→ Reference: **Example 4**

Solutions: [Solutions](#)

Practice 8: Real Battle

A 3×3 matrix M has column vectors $\vec{v}_1 = (2, 0, 0)$, $\vec{v}_2 = (0, 3, 0)$, $\vec{v}_3 = (0, 0, 5)$. Find $\det(M)$. What is the volume of the parallelepiped formed by these three vectors? If we apply M

to the unit cube, what shape results and what is its volume?

→ Reference: **Example 5, Visual Interlude**

Solutions: [Solutions](#)

Basic Algebra Drill — Matrices and Vectors (10 Problems)

Pure calculation. Build fluency with fundamental operations.

D1. Multiply $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} \times \begin{pmatrix} 4 & -1 \\ 2 & 1 \end{pmatrix}$.

D2. Compute the determinant of $\begin{pmatrix} 3 & 5 \\ 2 & 4 \end{pmatrix}$.

D3. Find $|\vec{a}|$ for $\vec{a} = (6, -8)$.

D4. Compute $\vec{a} \cdot \vec{b}$ for $\vec{a} = (1, -2, 3)$, $\vec{b} = (4, 0, -1)$.

D5. Add the matrices: $\begin{pmatrix} 1 & 3 \\ 2 & -1 \end{pmatrix} + \begin{pmatrix} 4 & 0 \\ -2 & 5 \end{pmatrix}$.

D6. Scale the vector: $5 \cdot (-2, 3, 1)$. Write as a vector.

D7. Compute the cross product: $(2, 1, 0) \times (0, 3, 1)$.

D8. Find a unit vector in the direction of $\vec{v} = (3, -4)$.

D9. Compute $\det \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$.

D10. Find the dot product of $\vec{u} = (2, -1)$ and $\vec{v} = (-3, 4)$. Are they perpendicular?

Solutions: [Solutions](#)

Advanced Algebra Drill — Matrices and Vectors (10 Problems)

Multi-step. Connect matrices to transformations and vectors to geometry.

- A1.** Find A^{-1} for $A = \begin{pmatrix} 4 & 3 \\ 3 & 2 \end{pmatrix}$. Then compute $A \times A^{-1}$ to verify.
- A2.** Find a vector perpendicular to both $\vec{a} = (1, 2, 3)$ and $\vec{b} = (4, 5, 6)$ using the cross product.
- A3.** Write the rotation matrix for 60° counterclockwise. Apply it to the vector $(1, 0)$. What are the new coordinates?
- A4.** Solve the system: $2x + y - z = 3$, $x - y + 2z = 1$, $3x + 2y + z = 7$.
- A5.** Find the angle between $\vec{a} = (1, 1, 0)$ and $\vec{b} = (0, 1, 1)$.
- A6.** Find the scalar and vector projections of $\vec{a} = (5, 12)$ onto $\vec{b} = (3, 4)$.
- A7.** Compute A^2 for $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.
- A8.** Find the area of the triangle with vertices $(0, 0, 0)$, $(2, 1, 0)$, $(0, 3, 0)$ using the cross product.
- A9.** Find a 2×2 matrix A such that $A^2 = I$ but $A \neq \pm I$. (Hint: think about reflections.)
- A10.** The columns of a 2×2 matrix are $(3, 1)$ and $(1, 3)$. Find the area of the parallelogram they span.

Solutions: [Solutions](#)

Today's Procedure

Step 1: Matrix operations – add entry-by-entry, scale entry-by-entry.
Multiply: row $i \times$ column j . Check inner dimensions match.
Inverse (2×2): swap a,d ; negate b,c ; divide by $\det$.
Determinant = area scaling in 2D, volume scaling in 3D.

Step 2: Vector operations – magnitude: $\sqrt{\text{sum of squares}}$.
Dot product: multiply components, sum $\rightarrow \cos \theta \rightarrow$ angle.
Cross product (3D): formula gives perpendicular vector. $|\mathbf{a} \times \mathbf{b}| = \text{area}$.
Projection: $(\mathbf{a} \cdot \mathbf{b} / |\mathbf{b}|^2) \mathbf{b}$ gives the shadow of $\mathbf{a}$ onto $\mathbf{b}$.

Step 3: Connection – matrix $\times$ vector = new vector. Columns of a matrix are the images of the standard basis vectors. In any dimension, a matrix transforms space; the determinant tells you the volume scale.

How to Read These Symbols

Symbol	Reads as	Meaning
$\vec{v} = \langle v_1, v_2 \rangle$	"vector v equals angle-bracket v_1 comma v_2 "	vector in component form
$ \vec{v} $	"magnitude of v " / "norm of v " / "length"	$\sqrt{v_1^2 + v_2^2}$ — length of the arrow
$\vec{v} \cdot \vec{w}$	" v dot w " / "dot product"	$v_1 w_1 + v_2 w_2$ — scalar result, related to cosine of angle
$\vec{v} \cdot \vec{w} = \vec{v} \vec{w} \cos \theta$	" v dot w equals norm v times norm w times cosine theta"	geometric dot product formula
$A = [a_{ij}]$	" A equals matrix with entries a_{ij} "	matrix: i =row index, j =column index
AB	" A times B " / "matrix product"	$(AB)_{ij} = \sum_k a_{ik} b_{kj}$ — row of A dot column of B

Symbol	Reads as	Meaning
A^{-1}	"A inverse"	matrix inverse: $AA^{-1} = A^{-1}A = I$
I	"I" / "the identity matrix"	ones on diagonal, zeros elsewhere — like multiplying by 1
$\det(A)$	"determinant of A"	$\det = ad - bc$ for 2×2 — zero means singular (no inverse)
A^T	"A transpose"	rows become columns, columns become rows
orthogonal	"orthogonal"	$\vec{v} \cdot \vec{w} = 0$ — perpendicular vectors
$\vec{0}$	"zero vector"	vector with all components zero

Terminology

Up to now we used plain words like "grid", "dot product", "shadow", "cross".

You have already learned all the methods. Now we attach the formal mathematical names.

What we called it	Mathematical term	Notation
matrix	matrix	$A = (a_{ij})$
determinant	determinant	$\det(A) = ad - bc$
identity matrix	identity matrix	$I, AI = IA = A$
inverse matrix	matrix inverse	$A^{-1}, AA^{-1} = I$
dot product	dot product / scalar product	$\vec{a} \cdot \vec{b}$
cross product	cross product / vector product	$\vec{a} \times \vec{b}$
projection	vector projection	$\text{proj}_{\vec{b}} \vec{a}$

What we called it	Mathematical term	Notation
linear transformation	linear transformation	$\vec{x} \mapsto A\vec{x}$
basis vector	standard basis vector	$\vec{e}_1 = (1, 0), \vec{e}_2 = (0, 1)$
parallelepiped	parallelepiped	3D analog of parallelogram
hyper-volume	hyper-volume / n -dimensional volume	\$